IC Tester Using ATmega32A and Dual 74HC4051

Mode-Selectable Firmware via UART (CodeVisionAVR)

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1. Project Overview

This project implements an 8-pin IC test platform using an ATmega32A microcontroller and two 74HC4051 8-channel analog multiplexers. The firmware supports two operating modes selectable through UART:

- Mode 1: Single MUX scan (measurement only, injection disabled)
- Mode 2: Dual MUX matrix test (injection + measurement), including automatic ground-pin detection. The design focuses on stable ADC readings using averaging, dummy conversions after MUX switching, and outlier rejection in the dual-mode measurement path.

2. System Requirements

2.1 Hardware

Item	Quantity	Notes
ATmega32A	1	External crystal at 11.0592 MHz
74HC4051 analog MUX	2	MUX1 = injection, MUX2 = measurement
IC socket / fixture	1	8-pin IC under test
UART interface	1	USB-UART adapter (9600 baud)
5V supply	1	Stable VCC and AVCC recommended
Decoupling capacitors	Recommended	0.1 uF near MCU and each MUX

2.2 Software

- CodeVisionAVR toolchain
- Serial terminal (e.g., PuTTY, Tera Term) configured for 9600 baud, 8 data bits, no parity, 1 stop bit (8N1)
  - Optional: AVR programmer for flashing ATmega32A

3. Hardware Design

3.1 MUX Roles

MUX1 (Injection): Selects one IC pin to receive VCC (+5V) through the MUX COM pin.

MUX2 (Measurement): Selects one IC pin at a time and routes it to the MCU ADC0 input via the MUX COM pin.
The MUX1 INH pin is controlled by the MCU to enable or disable injection safely.

3.2 Power and Reference Notes

The ADC reference is AVCC (ADMUX = 0x40). For best accuracy:

- Ensure AVCC is properly connected and decoupled.
- Use a stable 5V supply.
- If measured AVCC differs from 5000 mV, update the firmware constant VCC_MV accordingly.

4. Firmware Design

4.1 Startup Behavior and Safety

On boot, the firmware disables injection by default (MUX1 INH = 1) to avoid unintended powering of the IC. The UART menu is then printed and the firmware waits for user input ('1' or '2') before running a test.

4.2 UART Interface

UART configuration:

- Baud rate: 9600 (UBRR = 71 at 11.0592 MHz)
- Frame: 8N1

The user selects the test mode by sending '1' or '2'.

4.3 ADC Strategy

The firmware uses multiple techniques to reduce noise and improve repeatability:

- ADC warm-up conversions at startup
- Dummy ADC reads after each MUX channel change (flush sample/hold)
- Multi-sample averaging in Mode 1
- In Mode 2: large sample buffers, sorting, and trimming top/bottom outliers

5. Operating Modes

5.1 Mode 1 - Single MUX Scan (Measurement Only)

Purpose: Scan all 8 channels on MUX2 and report raw ADC and converted millivolts without injecting VCC.

Key characteristics:

- MUX1 disabled (INH = 1)
- Short settling time per channel
- Average of AVG_SAMPLES_SINGLE readings per channel

Mode 1 key parameters (from firmware):

Parameter	Value	Meaning

SETTLE_MS_SINGLE	10 ms	Wait after switching MUX2 channel
SAMPLE_DELAY_MS_SINGLE	2 ms	Delay between ADC samples
AVG_SAMPLES_SINGLE	20	Number of samples averaged per channel
VCC_MV	5000 mV	Used for raw-to-mV conversion (update if needed)

5.2 Mode 2 - Dual MUX Matrix Test (Injection + Measurement)

Purpose: For each injected pin (selected via MUX1), measure the voltages of all pins (via MUX2) to build an 8x8 measurement matrix. The pin whose injected row has the lowest average response is reported as the ground pin.

Key characteristics:

- MUX1 enabled (INH = 0)
- MUX1 channel steps 0..7, and for each, MUX2 scans 0..7
- Each measurement uses outlier rejection and averaging
- Results are normalized for display

Mode 2 key parameters (from firmware):

Parameter	Value	Meaning
NUM_SAMPLES	70	Number of ADC samples collected per reading
SETTLE_TIME	60 ms	Settling delay after changing a MUX channel
SAMPLE_DELAY	4 ms	Delay between samples inside a reading
NUM_MEASUREMENT_PASSES	2	Repetition count per matrix cell

6. Measurement and Filtering Details

6.1 Outlier Rejection in Mode 2

In Mode 2, each cell measurement is created by sampling NUM_SAMPLES ADC readings, converting them to volts, sorting the values, and discarding the top and bottom 15%. The remaining samples are averaged to produce a robust voltage estimate that is less sensitive to switching spikes and noise.

6.2 Multi-Pass Averaging

Each matrix cell is measured multiple times (NUM_MEASUREMENT_PASSES). The firmware averages these passes to further improve repeatability.

7. Ground Pin Detection

7.1 Selection Criterion

After building the 8x8 measurement matrix, the firmware computes an average for each injected row. The row with the lowest average voltage is selected as the ground pin candidate:

groundPin = argmin(channelAverages[i])

7.2 Normalization for Display

To display the results clearly, the selected row (groundPin) is normalized between 0 and 1 using its minimum and maximum values:

normalized[i] = (value[i] - min) / (max - min)

If the range is near zero, all normalized values are reported as 0.

8. How to Use the System

1) Connect the hardware.
2) Flash the firmware to ATmega32A.
3) Open a serial terminal at 9600 baud, 8N1.
4) Reset the MCU. The mode menu will appear.
5) Send '1' for single scan or '2' for dual matrix test.
6) Read the results from the terminal. The firmware halts at the end; press RESET to run again.

9. Configuration and Calibration

- VCC calibration: If AVCC is not exactly 5.000 V, measure it and update VCC_MV.
- Settling delays: If readings are unstable due to wiring length or IC load, increase SETTLE_TIME (Mode 2) and/or SETTLE_MS_SINGLE (Mode 1).
- Sampling: Increasing AVG_SAMPLES_SINGLE or NUM_SAMPLES improves stability but increases runtime.

10. Validation Checklist

Recommended checks before using the tester on unknown ICs:

- Verify PC4 controls MUX1 INH correctly (0 enables injection, 1 disables injection).
- Confirm ADC0 sees expected voltages when connecting a known divider to MUX2.
- Confirm MUX select lines PD2..PD7 match the intended channels.

• Confirm the IC socket mapping matches MUX channel numbering (0..7).

11. Limitations and Future Work

Limitations:

- Designed for 8-pin ICs (one-to-one mapping to MUX channels)
- Provides electrical signature and ground detection, not full functional IC identification • Results depend on stable wiring and supply quality

Future improvements:

- Add automatic logging to CSV on a connected PC
- Add support for 14/16-pin ICs using additional MUX devices
- Add signature matching database for known ICs
- Add protection circuitry (current limiting, clamp diodes) for unknown devices

Appendix A - Key Firmware Constants

```
// Clock: 11.059200 MHz
#define VCC_MV 5000L

// Mode 1 (single scan)
#define SETTLE_MS_SINGLE        10
#define SAMPLE_DELAY_MS_SINGLE   2
#define AVG_SAMPLES_SINGLE      20

// Mode 2 (dual scan)
#define NUM_SAMPLES 70
#define SETTLE_TIME 60
#define SAMPLE_DELAY 4
#define NUM_MEASUREMENT_PASSES 2
```

Appendix B - Mode Menu Output (UART)

```
=== IC Tester (Mode Select) ===
ATmega32A @ 11.059200 MHz
PC4 -> MUX1 INH (0=EN, 1=DIS)
 Choose
mode:
1) Single MUX scan (MUX2->ADC0, MUX1 disabled)
2) Dual MUX matrix test (MUX1 inject + MUX2 measure) Enter 1
   or 2:
```

12. Conclusion

This project provides a **reliable, noise-resistant IC testing platform** using simple hardware and advanced firmware techniques.
The dual-mode design allows both **safe passive measurement** and **active functional analysis**, making it suitable for:

- IC reverse engineering
- Educational labs
- Electronics debugging tools